

This test is a closed-book and closed-notes exam. No electronic devices or calculators are allowed.

All of your solutions should be easily identifiable and supporting work must be shown. Please clearly label what work you want to be considered for grading. Ambiguous or illegible answers will not be counted as correct.

The duration of the exam is 50 minutes. If you finish early, feel free to submit your exam and leave the room, but not in the last 5 minutes.

Name: Exam Solutions

Student ID Number: _____

Signature: _____

I certify, on my honor, that I have neither given nor received any help, or used any non-permitted resources while completing this evaluation.

1. (25 points) **Unique Interpolants and Degrees of Freedom:** Determine when interpolating polynomials and interpolating splines are unique and when uniqueness is subject to degree of freedom choices.

(a) (10 points) Given $n + 1$ points $\{x_i\}_{i=0}^n$ and a function evaluated at those points $\{f(x_i)\}_{i=0}^n$

- Define an interpolating polynomial $p_n(x)$ using a *monomial basis*. State the conditions satisfied by the polynomial at each point x_i .

$$p_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

where $p_n(x_i) = f(x_i) \quad \forall i \in \{0, \dots, n\}$.

- Define an interpolating polynomial $p_n(x)$ using the *Lagrange polynomial basis*. State the conditions satisfied by the polynomial at each point x_i .

$$p_n(x) = \sum_{i=0}^n L_{n,i}(x) f(x_i) \quad \text{where } L_{n,i}(x) = \frac{(x-x_0) \dots (x-x_{i-1})(x-x_{i+1}) \dots (x-x_n)}{(x_i-x_0) \dots (x_i-x_{i-1})(x_i-x_{i+1}) \dots (x_i-x_n)}$$

st $p_n(x_i) = f(x_i) \quad \forall i \in \{0, \dots, n\}$

- Define a degree k interpolating spline $s_k(x)$ using a *left monomial basis* $((x - x_i)$ on the interval $x_i \leq x \leq x_{i+1}$). State the conditions satisfied by the spline at each point x_i .

$$s_n(x) = a_i^k (x-x_i)^k + b_i^{k-1} (x-x_i)^{k-1} + \dots + c_i (x-x_i) + d_i \quad \text{on each } [x_i, x_{i+1}]$$

with $s_n(x_i) = f(x_i) \quad \forall i \in \{0, \dots, n\}$

$$\lim_{x \rightarrow x_i^-} s_n^{(r)}(x) = \lim_{x \rightarrow x_i^+} s_n^{(r)}(x) \quad \forall i \in \{1, \dots, n-1\}$$

(b) (5 points) Let the collected data be $f(0) = 0$, $f(1) = 1$, $f(2) = 4$, $f(3) = 9$.

- Write $p_3(x)$ in the Lagrange polynomial basis and the monomial basis. Then find $p_3(\frac{3}{2})$.

$$L_{n,0} = \frac{(x-1)(x-2)(x-3)}{(0-1)(0-2)(0-3)} = -\frac{(x-1)(x-2)(x-3)}{6} = -\frac{(x-1)}{6} \cdot x^2 - 5x + 6 = -\frac{1}{6}(x^3 - 5x^2 + 6x)$$

$$L_{n,1} = \frac{(x-0)(x-2)(x-3)}{(1-0)(1-2)(1-3)} = +\frac{x(x-2)(x-3)}{2} = \frac{x(x^2 - 5x + 6)}{2} = \frac{1}{2}(x^3 - 5x^2 + 6x)$$

$$L_{n,2} = \frac{(x-0)(x-1)(x-3)}{(2-0)(2-1)(2-3)} = -\frac{x(x-1)(x-3)}{2} = -\frac{x(x^2 - 4x + 3)}{2} = -\frac{1}{2}(x^3 - 4x^2 + 3x)$$

$$L_{n,3} = \frac{(x-0)(x-1)(x-2)}{(3-0)(3-1)(3-2)} = +\frac{x(x-1)(x-2)}{6} = \frac{1}{6}x(x^2 - 3x + 2) = \frac{1}{6}(x^3 - 3x^2 + 2x)$$

$$p_3(x) = \cancel{0} + \frac{1}{2}(x^3 - 5x^2 + 6x) - 2(x^3 - 4x^2 + 3x) + \frac{9}{6}(x^3 - 3x^2 + 2x)$$

$$= \left(\frac{1}{2} - 2 + \frac{3}{2}\right)x^3 + \left(-\frac{5}{2} + \frac{16}{2} - \frac{9}{2}\right)x^2 + \left(\frac{6}{2} - \frac{12}{2} + \frac{6}{2}\right)x = \boxed{x^2}$$

$p(\frac{3}{2}) = \boxed{\frac{9}{4}}$

- (c) (2 points) State whether the degree n interpolating polynomial is unique for $n+1$ many data points. Justify your answer by establishing the degrees of freedom in an interpolating polynomial.

Yes. An n degree poly $p_n(x) = a_n x^n + \dots + a_0$
 has $n+1$ degrees of freedom $a_0, \dots, a_n$.
 Provided $n+1$ many unique data points then there is a
 unique soln.

- (d) (3 points) State whether the cubic interpolating spline is unique on $n+1$ data points. Justify your answer by establishing the degrees of freedom in an interpolating spline.

No. A cubic spline has $4n$ degrees of freedom
 and $4n-2$ conditions, provided $n+1$ many data points.

- (e) (5 points) Using exact values, state *two different choices* for the degrees of freedom that will ensure a cubic spline interpolant will identically be a quadratic polynomial for the data provided in part (b).

$$s'(0)=0 \text{ \& } s'(3)=6 \quad \text{or} \quad s''(0)=2 \text{ \& } s''(3)=2$$

↑

Notice natural bc
 will not result in a
 quadratic

2. (25 points) **Lagrange Bases Under Integration and Differentiation:** The goal of this problem is to study how the Lagrange interpolating basis changes when we approximate function *integrals* and *derivatives*.

(a) (10 points) Consider the three equally spaced nodes $x_{-1} = -1$, $x_0 = 0$, $x_1 = 1$ and their Lagrange basis polynomials $\{\ell_{-1}(x), \ell_0(x), \ell_1(x)\}$.

(i) Write explicit formulas for $\ell_{-1}(x)$, $\ell_0(x)$, $\ell_1(x)$.

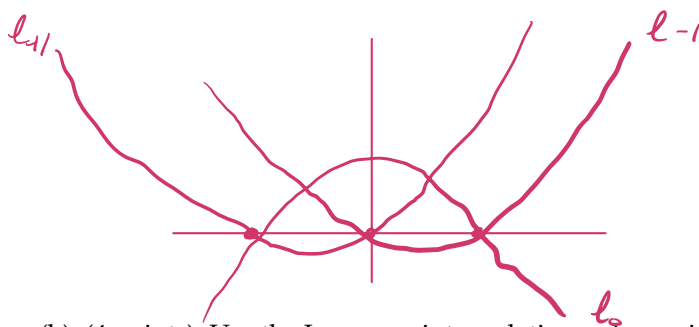
$$\ell_{-1}(x) = \frac{(x-0)(x-1)}{(-1-0)(-1-1)} = \frac{1}{2}(x^2 - x) \quad \ell_1(x) = \frac{(x-0)(x+1)}{(1-0)(1+1)} = \frac{1}{2}(x^2 + x)$$

$$\ell_0(x) = \frac{(x+1)(x-1)}{(0+1)(0-1)} = -(x^2 - 1)$$

(ii) Verify the partition of unity property $\ell_{-1}(x) + \ell_0(x) + \ell_1(x) \equiv 1$.

$$-x^2 + 1 + \frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{2}x^2 + \frac{1}{2}x = 1.$$

(iii) On the interval $[-1, 1]$, sketch the three basis polynomials on the same axes. Label roughly where each is positive/negative and indicate their values at $x = -1, 0, 1$.



(b) (4 points) Use the Lagrange interpolating polynomial $p(x)$ to estimate the integral. Compute

$$I_{-1} = \int_{-1}^1 \ell_{-1}(x) dx, \quad I_0 = \int_{-1}^1 \ell_0(x) dx, \quad I_1 = \int_{-1}^1 \ell_1(x) dx.$$

and show that

$$\int_{-1}^1 p(x) dx = \frac{1}{3}(f(-1) + 4f(0) + f(1)).$$

$$\int_{-1}^1 p(x) dx = \sum_{i=0}^n f(x_i) \int_{-1}^1 \ell_i(x) dx$$

$$\int_{-1}^1 \ell_{-1}(x) dx = \left[\frac{1}{6}x^3 - \frac{1}{4}x^2 \right]_{-1}^1 = \frac{1}{6} \cdot 2 = \frac{1}{3}$$

$$\int_{-1}^1 \ell_0(x) dx = \left[-\frac{1}{3}x^3 + x \right]_{-1}^1 = 2 \cdot \frac{2}{3} = \frac{4}{3}$$

$$\int_{-1}^1 \ell_1(x) dx = \left[\frac{1}{6}x^3 + \frac{1}{4}x^2 \right]_{-1}^1 = 2 \cdot \frac{1}{6} = \frac{1}{3}$$

$$\int_{-1}^1 p(x) dx = \frac{1}{3}[f(-1) + 4f(0) + f(1)].$$

- (c) (4 points) Use the Lagrange interpolating polynomial $p(x)$ to estimate the derivative at $x = 0$. Compute $\ell'_{-1}(0)$, $\ell'_0(0)$, $\ell'_1(0)$ and show that

$$p'(0) = \sum_{i=0}^n f(x_i) \ell'_i(0)$$

$$p'(0) = \frac{f(1) - f(-1)}{2}$$

$$\ell'_{-1}(x) = x - \frac{1}{2}$$

$$\ell'_0(x) = -2x$$

$$\ell'_1(x) = x + \frac{1}{2}$$

$$p'(0) = \left(\frac{1}{2}\right)f(1) + 0 \cdot f(0) + \left(-\frac{1}{2}\right)f(-1)$$

$$= \frac{f(1) - f(-1)}{2}$$

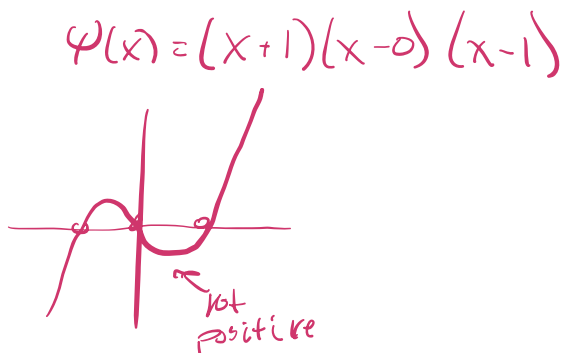
- (d) (4 points) A student wants to analyze the quadrature error

$$\int_{-1}^1 (f(x) - p(x)) dx = \int_{-1}^1 \frac{f^{(3)}(\xi_x)}{3!} \Psi(x) dx$$

by claiming that, by the Integral Mean Value Theorem, there exists $c \in (-1, 1)$ such that

$$\int_{-1}^1 \frac{f^{(3)}(\xi_x)}{3!} \Psi(x) dx = \frac{f^{(3)}(c)}{3!} \int_{-1}^1 \Psi(x) dx.$$

Is this reasoning valid in general? Justify your answer carefully, and support your explanation with a sketch of $\Psi(x)$ on $[-1, 1]$.



The argument relies on $\Psi(x)$ being non-negative over the interval. But $\Psi(x)$ is negative on $(-1, 0)$.

- (e) (3 points) Suggest an alternative approach using the function $w(x) = \int_{-1}^x (t-1)t(t+1)dt$ and how it impacts the order of the error. You only need to reason about the approach. You do not need to find the solution.

An alternative apch involves considering

$$\int_{-1}^1 w'(x) \frac{f^{(3)}(\xi_x)}{3!} dx \quad \text{and using integration}$$

by parts to raise the order of the error.

eg. this is the reason Simpson's is much more accurate than trapezoidal

Feel free to use this as scratch.